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#31 (1571308266): HemaVision-Net: A High-Resolution Convolutional Framework for Severely Occluded Blood Smears

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	Drag to change order	Author name	Author affiliation (edit for paper)	Author email	Email	Delete
Authors		Meheraj Hasan	American International University- Bangladesh, Bangladesh	22-49363-3@student.aiub.edu		
		Nazifa Tabushum Turaba	American International University- Bangladesh, Bangladesh	22-46271-1@student.aiub.edu		
		Ainea Esrat Esika	American International University- Bangladesh, Bangladesh	22-46837-1@student.aiub.edu		
		Sanjid Raaz	American International University- Bangladesh, Bangladesh	21-44896-2@student.aiub.edu		
		Sultanul Arifeen Hamim	American International University- Bangladesh, Bangladesh	hamim@aiub.edu		
		Muhammad Firoz Mridha	American International University- Bangladesh, Bangladesh	firoz.mridha@aiub.edu		

Full paper title	HemaVision-Net: A High-Resolution Convolutional Framework for Severely Occluded Blood Smears		
Conference and track	2026 7th International Conference on Data Analytics for Business and Industry (ICDABI) - 15 Image Analysis & Tomography		
Abstract	The peripheral blood smear (PBS) analysis plays a crucial role in diagnosing hematological...		
Similarity	On Sep 2, ithenticate computed a similarity score of 5 for the review manuscript.		
Personal notes			
Roles	You are the creator and an author for this paper. You have authored an accepted paper in this conference.		
Status	Accepted with major revisions		
Copyright			
Presented	not specified		
Review manuscript	Final Manuscript	Revised Paper	Revised Manuscript (with highlighted changes)

Full Review

Rebuttal

Full Review

Originality	Significance of Topic	Presentation
Weak Accept6	Accept8	Weak Accept6

Strengths/Weakness

Strengths: The paper addresses an important and challenging problem in automated blood smear analysis, particularly the detection of heavily overlapping RBCs. The proposed high-resolution YOLOv11-based framework, two-stage training strategy, and comparison with RT-DETR and SAHI provide a clear technical direction. The model achieves 92.93% mAP@0.5 on the internal BCCD test set and 84.99% on an unseen external dataset.

Weaknesses: The main concerns are the limited external validation scope, incomplete computational efficiency analysis, and some methodological/technical inconsistencies in the reported evaluation metrics. The claim of clinical applicability is stronger than what can be concluded

Contribution/s & Detailed comments

The paper proposes HemaVision-Net, a high-resolution CNN framework based on YOLOv11 for detecting RBCs, WBCs, and platelets in densely populated and severely occluded blood smear images. The main contributions include a two-stage training strategy, high-resolution fine-tuning, architectural comparison with RT-DETR, and analysis of SAHI-based inference.

The following issues should be addressed:

1. Clarify the novelty of HemaVision-Net. Since the architecture is based on native YOLOv11 components such as C2f and PANet, the authors should clearly identify which architectural modifications constitute the actual methodological contribution.

from the current experiments.

2. Strengthen the external validation. The external validation currently uses one WBC Object Detection Dataset and reports 84.99% mAP@0.5. A stronger cross-institutional evaluation involving multiple independent datasets would provide more convincing evidence of generalization.

3. Provide computational efficiency analysis. The paper acknowledges that 800×800 inference increases VRAM requirements, but it does not report inference time, FPS, GPU/CPU memory consumption, parameter count, or FLOPs. These metrics are important for assessing the practical deployment claims.

4. Clarify the ablation design. The ablation study compares RT-DETR and HemaVision-Net and evaluates high-resolution and SAHI inference, but it does not isolate the contribution of individual modifications such as the increased box-loss weight, two-stage pretraining, input resolution, and MixUp removal. A component-level ablation would better establish which modifications are responsible for the performance improvement.

5. Correct the recall formulation. Equation (3) defines recall as $TP/(TP + TN)$, whereas the standard recall definition is $TP/(TP + FN)$. This is an important technical error and should be corrected.

6. Clarify the role of TTA. The manuscript states that 800-pixel inference combined with Test-Time Augmentation preserves 92.93% mAP, but Table II labels the corresponding configuration simply as “HemaVision-Net Standard (800px).” The relationship between standard inference and TTA should be explicitly clarified.

7. Moderate the clinical applicability claims. The study demonstrates promising detection performance, but the experiments do not yet establish clinical diagnostic validity. The current task is detection of three cell categories rather than diagnosis of hematological abnormalities. The authors themselves acknowledge this limitation.

8. Improve class-level error analysis. RBC detection has an mAP@0.5 of 89.2% and precision of only 73.9%, compared with 99.3% mAP for WBCs. The paper should provide more detailed false-positive and false-negative analysis for severely occluded RBCs, including representative failure cases.

9. Clarify the “SAHI Paradox” terminology. The observed degradation of performance with SAHI is interesting, but the term should be presented as an empirical observation of this study rather than as a broadly established phenomenon. Additional experiments with different slice sizes, overlaps, and NMM configurations would strengthen this conclusion.

10. Improve reproducibility. Important details such as dataset sample counts after preprocessing, exact training configuration, hardware used, random seeds, and software/library versions should be provided so that the reported results can be independently reproduced.

Strengths/Weakness

- The paper addresses an important medical image analysis problem, particularly detection of densely packed and overlapping blood cells where conventional object detection can be challenging.
- The combination of high-resolution YOLOv11, two-stage training, domain adaptation, and comparison with RT-DETR and SAHI provides a reasonably structured experimental study.
- The reported internal mAP@0.5 of 92.93% and external result of 84.99% indicate promising detection performance.
- The methodological novelty is limited because the proposed model is essentially a customized YOLOv11 configuration with modified resolution, augmentation, and loss weighting rather than a substantially new detection architecture.
- There are important technical inconsistencies in the evaluation methodology that need correction before the results can be considered fully reliable.

Contribution/s & Detailed comments

- The two-stage training strategy using TXL-PBC/CBC for pretraining and BCCD for fine-tuning is a reasonable domain-adaptation strategy and is clearly described.
- The ablation study comparing RT-DETR, high-resolution inference, standard HemaVision-Net, and SAHI provides useful evidence regarding the effect of inference strategy.
- The claimed "SAHI Paradox" is potentially interesting, but the conclusion should be stated more cautiously because the observed degradation may depend on slice size, overlap ratio, NMS/NMM configuration, and dataset characteristics.
- A major technical issue is the definition of Recall in Equation (3), which is given as $(TP/(TP+TN))$. The standard recall definition is $(TP/(TP+FN))$; this error should be corrected and all associated reported metrics should be rechecked.
- The manuscript should provide complete experimental details, including exact dataset sizes, class distributions, train/validation/test counts, preprocessing, random seeds, training epochs, optimizer, learning rate, batch size, and hardware configuration.
- The external validation claim should be interpreted carefully because the external WBC Detection Dataset is described as an unseen dataset, but the paper does not demonstrate equivalent external validation for all three target classes (RBC, WBC, and platelets).
- The claim that the model is suitable for "real clinical applicability" is too strong given that the evaluation is limited to three cell categories and does not include pathological cell subtypes or clinical diagnostic outcomes. The authors themselves acknowledge this limitation.
- The comparison against RT-DETR is useful, but additional comparison with recent YOLO-based blood-cell detectors would provide a more appropriate state-of-the-art benchmark because the proposed method itself is based on YOLOv11.
- Computational cost should be reported, including inference latency, FPS, GPU/VRAM requirements, and model parameters, especially because the proposed 800×800 high-resolution inference increases computational requirements.

Neutral

5 Accept

8 Reject

2

Strengths/Weakness

Strengths:

1. The author addressed a relevant and practical problem of detecting RBCs, WBCs and platelets in crowded, overlapping peripheral blood smear images.
2. Combines a YOLO based detector with high-

Contribution/s & Detailed comments

Problem addressed and importance. The paper addresses automated detection and classification of RBCs, WBCs, and platelets in crowded peripheral blood smear (PBS) images, where overlapping cells make detection difficult. This is an important, practical

resolution inference, two-stage training, augmentation, and loss re-weighting into a coherent pipeline.

3. Includes cross-dataset evaluation, which gives some evidence of generalisation.

4. The observation that SAHI slicing degraded performance in this setting is potentially interesting and worth further study.

Weakness:

1. The architecture consists largely of standard YOLOv11 components. The authors need to identify clearly what is added, removed, or modified. This seems to be tuning rather than methodological advancement. So the novelty is a bit unclear.
2. The "ablation study" is a comparison with RT-DETR. The individual effects of the architectural changes are never isolated.
3. There are substantial language and technical editing is need. For example, Recall is defined as $TP/(TP+TN)$ instead of $TP/(TP+FN)$. Also, the CloU is labelled as (L_DFL) in the text.
4. The explanations for SAHI's degradation (edge truncation, NMM failure) and for λ_{box} separating merged cells are presented as established causes without evidence. The terms "external validation" and "clinical applicability" overstate what a second Kaggle dataset can demonstrate for a cell-detection task.
5. Only mAP@0.5 is reported. mAP@0.5:0.95 and per-class precision, recall, and AP are missing. It is unclear whether the headline 92.93% includes TTA, and therefore whether the baseline comparisons were made under equivalent conditions. Dataset splits and leakage controls are insufficiently described.

problem for automated haematology. The authors' focus on dense, overlapping cells, rather than clean images of isolated cells, is well motivated.

The authors propose HemaVision-Net, which combines a YOLO-based detector with high-resolution (800×800) inference, two-stage training (TXL-PBC/CBC pretraining, then BCCD fine-tuning), augmentation, an increased box-loss weight ($\lambda_{box} = 10$), and TTA. The architecture, however, consists largely of standard YOLOv11 components (Conv-BN-SiLU, C2f, SPPF, multi-scale heads). The paper does not identify which modules were added, removed, or modified. As presented, the novelty lies mainly in engineering and tuning an existing detector rather than in a new method. The SAHI finding is the most original observation, but it is not yet sufficiently analysed.

The technical depth is currently limited by missing experiments and by errors in the core methodology.

Comments to the authors:

1. Please provide an explicit comparison with YOLOv11, listing each modified module and the rationale for its use on crowded PBS images. If the contribution is mainly the training and inference pipeline, please frame it that way.
2. The comparison with RT-DETR is a baseline comparison, not an ablation. Please isolate the effect of each component: the architectural changes, two-stage pretraining, 800×800 input, augmentation, λ_{box} , and TTA.
3. $\lambda_{box} = 10$. Please explain how this value was selected and report results for several values. The claim that it separates merged RBC clusters requires quantitative or visual evidence.
4. CloU is also mislabelled as L_DFL. Please give a complete, consistent L_{total} with its weighting factors.
5. Recall is given as $TP/(TP+TN)$ instead of $TP/(TP+FN)$. Please correct this and confirm that the implementation and reported values used the correct definition.
6. Please clarify what the 92.93% mAP@0.5 represents, which baseline it improves on, and whether it includes TTA. If it does, please evaluate the baselines under the same conditions. Please also report mAP@0.5:0.95 and per-class precision, recall, and AP. Tie each class-level figure to its metric (e.g. "platelet mAP@0.5 of 90.3%") and replace "near-perfect" with actual values.
7. Please report image and cell counts, splits, preprocessing, and resolutions. State whether splitting was done at the patient level and how leakage across datasets was excluded.
8. Testing on a second Kaggle dataset demonstrates cross-dataset generalisation. It is not clinical external validation unless independence of patients, equipment, and staining can be shown. Since the task is cell detection, please avoid diagnostic claims.
9. The observed drop in mAP@0.5 (92.9%→84.0% for HemaVision-Net; 89.6%→84.3% for RT-DETR) is interesting. However, edge truncation, context loss,

and NMM failure are presented as established causes without supporting evidence. Please provide: a boundary error analysis; visual examples; slice, overlap, and NMM settings; results with alternative merging configurations. Please also limit the conclusion to the evaluated configuration.

10. The claim that local receptive fields help separate overlapping cells should be presented as a hypothesis unless it is supported by visualisations or controlled experiments.

11. Please also acknowledge limited dataset diversity, staining and acquisition variation, pathological cell variants, and the absence of disease-level evaluation.

12. The manuscript needs thorough proofreading.

Examples:

"highresolution" → "high-resolution"

"close-perfect" → "near-perfect"

"furthers confirms" → "further confirms"

"Right here the main barrier lies..." → "The primary challenge lies in dense cellular occlusion."

The problem is important and the direction is promising. However, the paper does not yet establish its methodological novelty, lacks component-level ablations, and contains errors in its evaluation, equations and have multiple typos. I encourage the authors to address these points in a future submission.

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